

Data Dictionary — SoilGrids Features

Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France

Source: SoilGrids v2.0 — ISRIC World Soil Information · **Resolution:** 250 m × 250 m
Access method: WCS raster download (not REST API) · **Coverage:** Centre-Val de Loire bounding box
License: CC BY 4.0 · **Rows:** 1,500 parcels · **Columns:** 45 (6 identifiers + 39 soil features)

1. Overview

SoilGrids is a global system of digital soil mapping that uses machine learning to predict soil properties at 250 m resolution from ~240,000 soil profiles and remote-sensing covariates. For this project, 13 soil properties were retrieved at three root-zone depths, giving **39 features per parcel**, joined to the 1,500 wheat parcel sample by spatial extraction.

Why SoilGrids and not French sources (BDAT, RMQS)? French soil databases (BDAT, RMQS) provide real laboratory measurements but at coarse aggregation (canton or 16 km grid). SoilGrids, although modelled, delivers parcel-level resolution (250 m) and the water-retention properties (FC, WP) that French sources do not. The two are complementary; SoilGrids is used here as the parcel-level base, optionally enriched later with BDAT for chemistry features.

2. Column dictionary

KEY identifier **RAW** carried from parcel sample **FEATURE** SoilGrids variable

Identifiers (6 columns)

Column	Type	Unit	Description
parcelle_uid KEY	text	—	Unique observation identifier (year_IDPARCEL). Primary join key with weather, NDVI and target.
year KEY	int	year	Harvest year (2020–2024).
ID_PARCEL RAW	int	—	RPG parcel identifier.
SURF_PARC RAW	float	hectares	Parcel area.
lat RAW	float	degrees	Latitude of parcel centroid (WGS84).
lon RAW	float	degrees	Longitude of parcel centroid (WGS84).

Soil features (39 columns: 13 properties × 3 depths)

Each property is provided at three root-zone depths (0–5cm, 5–15cm, 15–30cm). Naming convention: *property_depth* (e.g. *phh2o_0-5cm*).

Property	Unit (raw)	Real-world value	Description & agronomic relevance
phh2o FEATURE	pH × 10	÷ 10 → pH 0–14	Soil pH in water. Drives nutrient availability and microbial activity. Typical agricultural range: pH 5.5–8.
nitrogen FEATURE	cg/kg	÷ 100 → g/kg	Total nitrogen content. Key fertility indicator for wheat.
soc FEATURE	dg/kg	÷ 10 → g/kg	Soil organic carbon. Indicator of soil health and water-holding capacity.
clay FEATURE	g/kg	÷ 10 → %	Clay content (< 2 µm particles). Controls water retention and cation exchange.
sand FEATURE	g/kg	÷ 10 → %	Sand content (50 µm – 2 mm). Controls drainage and aeration.
silt FEATURE	g/kg	÷ 10 → %	Silt content (2 – 50 µm). Texture component, intermediate water/aeration behaviour.
cec FEATURE	mmol(c)/kg	same	Cation Exchange Capacity. Soil's ability to retain nutrients (K ⁺ , Ca ²⁺ , Mg ²⁺).
bdod FEATURE	cg/cm ³ × 100	÷ 100 → g/cm ³	Bulk density of the fine earth. Indicator of compaction and porosity.
cfvo FEATURE	cm ³ /dm ³	÷ 10 → vol%	Coarse fragments volume (stones, gravel). Reduces effective soil volume.

wv0033	FEATURE	vol% × 10	÷ 10 → vol%	Field Capacity (water content at -33 kPa). Maximum plant-available water after drainage.
wv1500	FEATURE	vol% × 10	÷ 10 → vol%	Wilting Point (water content at -1500 kPa). Soil water at which plants wilt.
wv0010	FEATURE	vol% × 10	÷ 10 → vol%	Saturation water content (water at -10 kPa). Full pore-water condition.
ocd	FEATURE	hg/m³	÷ 10 → kg/m³	Organic carbon density. Carbon stock per unit volume.

Derived agronomic indicator (computable from these)

Indicator	Formula	Use
Available Water Content (AWC)	wv0033 – wv1500	Water actually usable by the crop; critical for stress modelling and wheat ripening.

3. Descriptive statistics (couche 0-5cm, after correction)

Variable	Count	Mean	Std	Min	Max	Real-world interpretation
phh2o_0-5cm	1500	65.8	3.4	52	72	pH 5.2–7.2 (slightly acidic to neutral)
nitrogen_0-5cm	1500	547	82	~300	~890	3–9 g/kg (normal cultivated soil)
clay_0-5cm	1500	250	44	~100	~400	10–40% clay (loamy soils)
sand_0-5cm	1500	266	110	~50	~570	5–57% sand (diversity captured)
wv0033_0-5cm	1500	314	26	~270	~390	27–39 vol% Field Capacity
wv1500_0-5cm	1500	157	21	~110	~215	11–21 vol% Wilting Point

All values are physically plausible and consistent with Centre-Val de Loire soil profiles (predominantly loamy with clay-loam in Beauce, sandier in Sologne, ouest Tourangeau).

4. Data quality and structure notes

- **No missing values** in the final dataset (1,500 × 45 fully populated).
- **2 parcels were initially nodata** at the raster pixel level (0.13%): both lay at the edge of soil-coverage zones. Corrected by **hybrid nearest-neighbour filling** — the closest valid pixel was used (mean interpolation distance: 0.22 km = 1 pixel).
- **Method documented:** this nearest-neighbour fill is the same approach used in the reference rice-yield thesis (Selena 2025) and in soil-mapping literature (McBratney et al., 2003).
- **Static data:** soil properties do not change between years. Although each parcel carries its year, the soil values reflect the parcel's location only.

5. Project usage

- **Join key:** `parcelle_uid` matches the parcel sample, the weather features (via department), and the Sentinel-2 series (to be added).
- **Suggested features for ML:** all 39 raw + derived AWC (wv0033 – wv1500) + texture ratios. Wheat is particularly sensitive to AWC during grain filling.
- **Most predictive expected features:** Field Capacity (wv0033), Wilting Point (wv1500), clay and sand content, organic carbon (soc).
- **Reproducibility:** rasters are downloaded once for the region's bounding box; any future re-sampling of parcels can reuse them.